

LOW ACRYLONITRILE NBR

1. ABSTRACT

This report provides an overview of the research scope, definition, and selected patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber). The target compound is defined as a nitrile rubber with low acrylonitrile content. The selected patent landscape includes patents related to the production of nitrile rubbers with varying acrylonitrile contents. The major polymerization approaches include emulsion polymerization and solution polymerization. The major formulation/process trends include the use of different monomer ratios, emulsifiers, initiators, and chain-transfer agents. The report aims to synthesize the extracted data from multiple patents to provide a comprehensive understanding of the production of Low Acrylonitrile NBR.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: EP4112676 - Patent Title: METHOD FOR PRODUCING NITRILE RUBBER - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2024-03-13 - Polymer Type: Nitrile Rubber - Relevance to target: The patent is relevant to Low Acrylonitrile NBR as it describes a method for producing nitrile rubber with varying acrylonitrile contents.

Polymerization / Synthesis Method - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent does not provide specific experimental evidence for the production of Low Acrylonitrile NBR.

Technical Relevance The patent is relevant to Low Acrylonitrile NBR as it describes a method for producing nitrile rubber with varying acrylonitrile contents. However, the patent does not provide specific details on the production of Low Acrylonitrile NBR.

Patent 2

Patent Details - Patent Number: US20160369014 - Patent Title: Process for the production of water and solvent-free nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2016-12-22 - Polymer Type: Nitrile Rubber - Relevance to target: The patent is relevant to Low Acrylonitrile NBR as it describes a process for the production of water and solvent-free nitrile rubbers.

Polymerization / Synthesis Method - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent provides examples of the production of nitrile rubbers with varying properties, including viscosity and residual water concentration. - The patent describes the use of different additives, such as calcium-stearate and antioxidants, to improve the properties of the nitrile rubbers.

Technical Relevance The patent is relevant to Low Acrylonitrile NBR as it describes a process for the production of water and solvent-free nitrile rubbers. The patent provides examples of the production of nitrile rubbers with varying properties and describes the use of different additives to improve the properties of the nitrile rubbers.

Patent 3

Patent Details - Patent Number: US3822242 - Patent Title: Manufacturing process for alternating copolymer of butadiene and acrylonitrile ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 1974-07-02 - Polymer Type: Nitrile Rubber - Relevance to target: The patent is

relevant to Low Acrylonitrile NBR as it describes a manufacturing process for the production of alternating copolymers of butadiene and acrylonitrile.

Polymerization / Synthesis Method - Polymerization process: Not disclosed in the analyzed patent evidence. - Monomer system: Not disclosed in the analyzed patent evidence. - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent provides examples of the production of alternating copolymers of butadiene and acrylonitrile with varying properties, including yield and intrinsic viscosity. - The patent describes the use of different catalyst systems and reaction conditions to improve the properties of the copolymers.

Technical Relevance The patent is relevant to Low Acrylonitrile NBR as it describes a manufacturing process for the production of alternating copolymers of butadiene and acrylonitrile. The patent provides examples of the production of copolymers with varying properties and describes the use of different catalyst systems and reaction conditions to improve the properties of the copolymers.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- The patents describe different methods for the production of nitrile rubbers, including emulsion polymerization and solution polymerization.
- The patents provide examples of the production of nitrile rubbers with varying properties, including viscosity, residual water concentration, and intrinsic viscosity.
- The patents describe the use of different additives, such as calcium-stearate and antioxidants, to improve the properties of the nitrile rubbers.
- The patents describe the use of different catalyst systems and reaction conditions to improve the properties of the copolymers.

4. REFERENCES

- [EP4112676](#)
- [US20160369014](#)
- [US3822242](#)